

Max Heap Sort (Heap Sort)

Definition

Heap Sort is a comparison-based sorting algorithm that uses a **heap data structure**. In **Max Heap Sort**, the largest element is kept at the root, then repeatedly moved to the end of the array.

Algorithm

1. Build a **Max Heap** from the given array.
2. The largest element is now at the root.
3. Swap the root with the last element.
4. Reduce the heap size by one.
5. Apply **heapify** to restore the Max Heap property.
6. Repeat until all elements are sorted.

OUTPUT:

```
Enter number of elements: 5
Enter element: 4
Enter element: 8
Enter element: 9
Enter element: 7
Enter element: 5
Original array: [4, 8, 9, 7, 5]
Max Heap: [9, 8, 4, 7, 5]
Time Complexity: O(n)
Space Complexity: O(log n)

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

Case	Complexity
Best Case	$O(n \log n)$
Average Case	$O(n \log n)$
Worst Case	$O(n \log n)$

Space Complexity

- **Auxiliary Space:** $O(1)$
- Heap Sort is an **in-place sorting algorithm**.